



MASTER OF SCIENCE IN MECHATRONIC SYSTEMS

Computer vision-based navigation of self-driving cars

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Introduction

- Computer vision-based navigation enables vehicles to perceive, understand, and navigate environments using cameras and sensors
 - Self-driving technology promises safer, more efficient transportation by removing human error from the equation
 - Advances in deep learning, sensor technology, and computing power have made autonomous navigation increasingly feasible
 - Real-world AV testing is costly, safety-critical, and difficult to reproduce under consistent conditions
 - Simulation enables safe, rapid development, but the reality gap causes performance degradation when deployed on hardware.
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Introduction - Aim & Objectives

➤ **Aim:** Develop and validate vision-based autonomous navigation on QLabS and QCar, quantifying sim-to-real performance

➤ **Objectives:**

- Dataset Preparation: Collect and annotate datasets from QLabS and QCar with augmentation
- Model Training: Train YOLOv8 for detection, compare U-Net/ENet for segmentation
- System Development: Integrate perception, AND-logic sensor fusion, and Pure Pursuit control
- Simulation Validation: Establish performance baselines in QLabS with documented protocols
- Hardware Deployment: Quantify reality gap and provide replicable framework for educational platforms



Literature Review

➤ Platform:

- QCar: 1:10 scale autonomous vehicle with NVIDIA Jetson TX2 (256 CUDA cores, 8GB RAM) (Quanser, n.d.a)
- Sensors: CSI cameras, Intel RealSense D435 depth camera and RPLiDAR A2
- Actuation: DC motor (throttle), Servo motor (steering), 720-count motor encoder with hardware digital tachometer
- QLab: Physics-based simulation with identical sensor API for simulation-to-hardware transfer (Quanser, n.d.b)

➤ Detection:

- CNNs preferred for embedded AV: fast, lightweight, easy to deploy with extensive pretrained options (Grigorescu et al., 2020)
- One-stage detectors predict boxes and classes in single forward pass, suitable for real-time control loops (Zou et al., 2023)
- YOLO architectures demonstrate real-time performance on Jetson-class embedded hardware (Ultralytics, 2023; Behrendt & Novak, 2017)

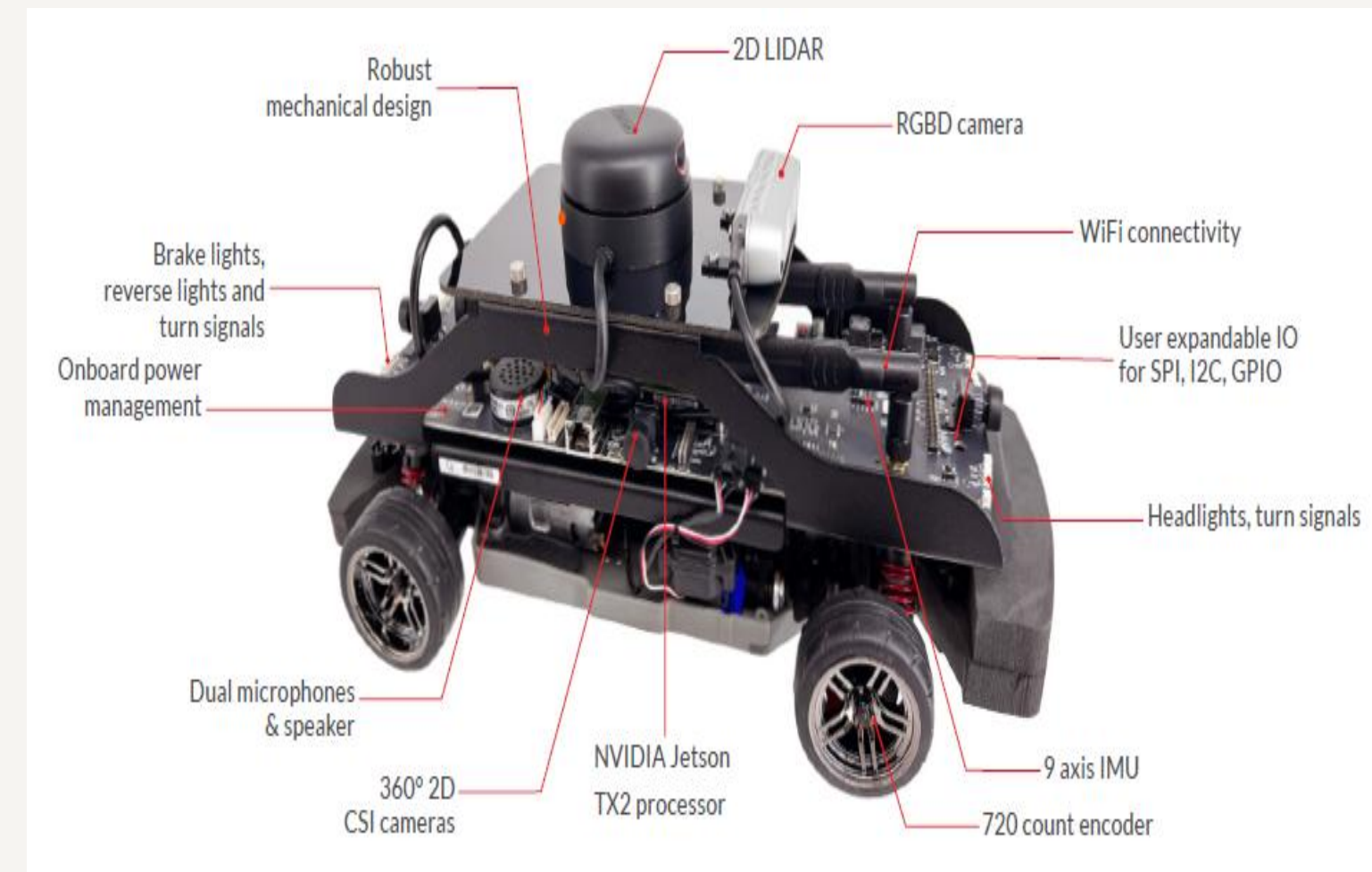


Fig 1. Quanser QCar technical specifications (Quanser, n.d.a)



Literature Review

➤ Segmentation:

- Semantic segmentation: classifies each pixel into categories (lanes, drivable surface, background) (Minaee et al., 2021)
- U-Net: skip connections preserve spatial details for precise boundary delineation, 31M parameters (Ronneberger et al., 2015)
- ENet: factorised convolutions enable real-time inference with 355K parameters (Paszke et al., 2016)

➤ Sensor Fusion:

- Depth sensors: accurate near-range but fail with some materials and limited range
 - LiDAR: reliable extended ranging with sparse sampling
 - Multi-modal perception combines complementary sensors to overcome individual limitations (Yeong et al., 2021)
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Methodology

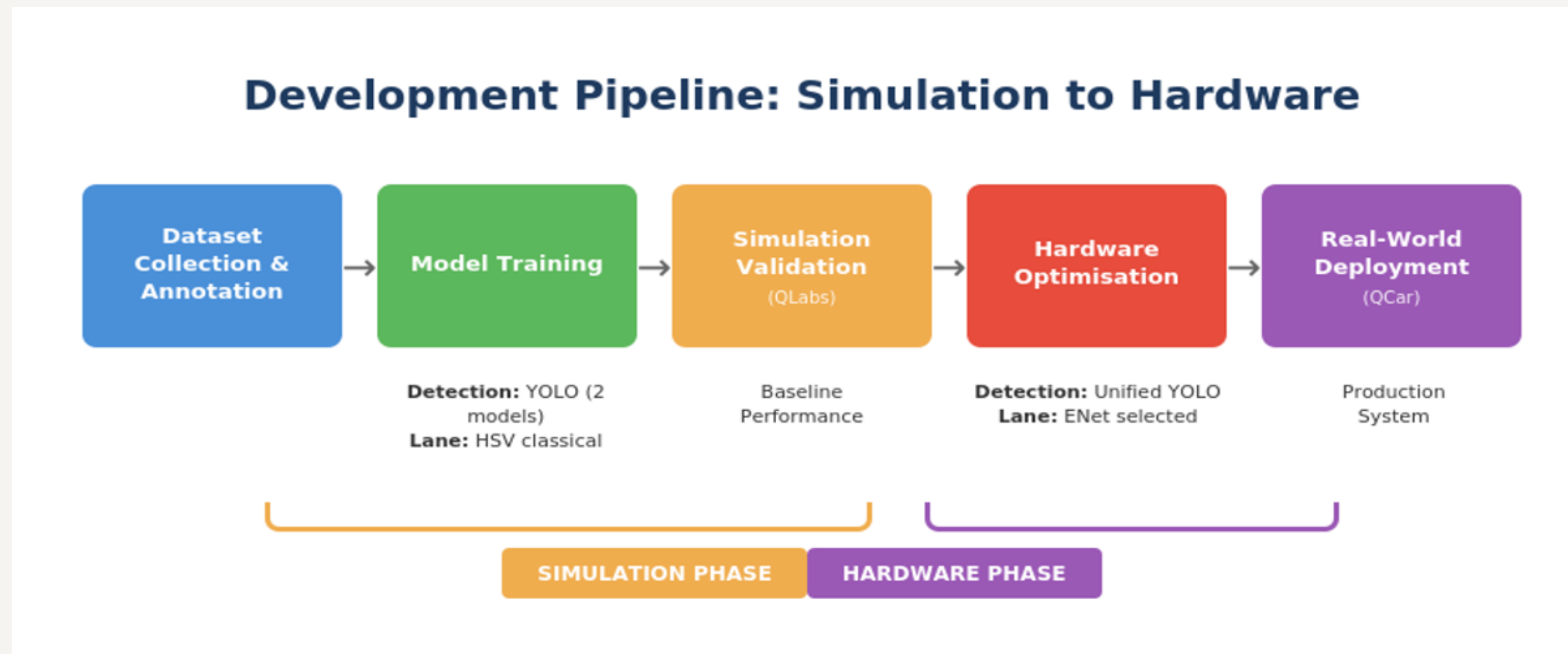


Fig. 2 Development pipeline

- Progressive development from QLABS simulation to physical QCar deployment
- System components: Perception, Control, and Sensor Fusion
- Simulation and hardware phases detailed in following slides

Methodology - Simulation Phase (QLabs)

- Q Labs offers consistent lighting and perfect geometry
 - for clean baseline development
- Object detection: Two separate YOLO models
 - for traffic lights and stop signs
- Traffic light dataset: 1,288 instances (red, green);
Stop sign dataset: 496 instances
- Lane following: HSV-based classical vision with colour thresholding and centroid extraction



Fig.3 Q Labs virtual environment

Methodology - Simulation Phase (QLabs)

- Dataset: 1,784 annotated instances for detection training
- Monolithic architecture: single Python process for all perception and control

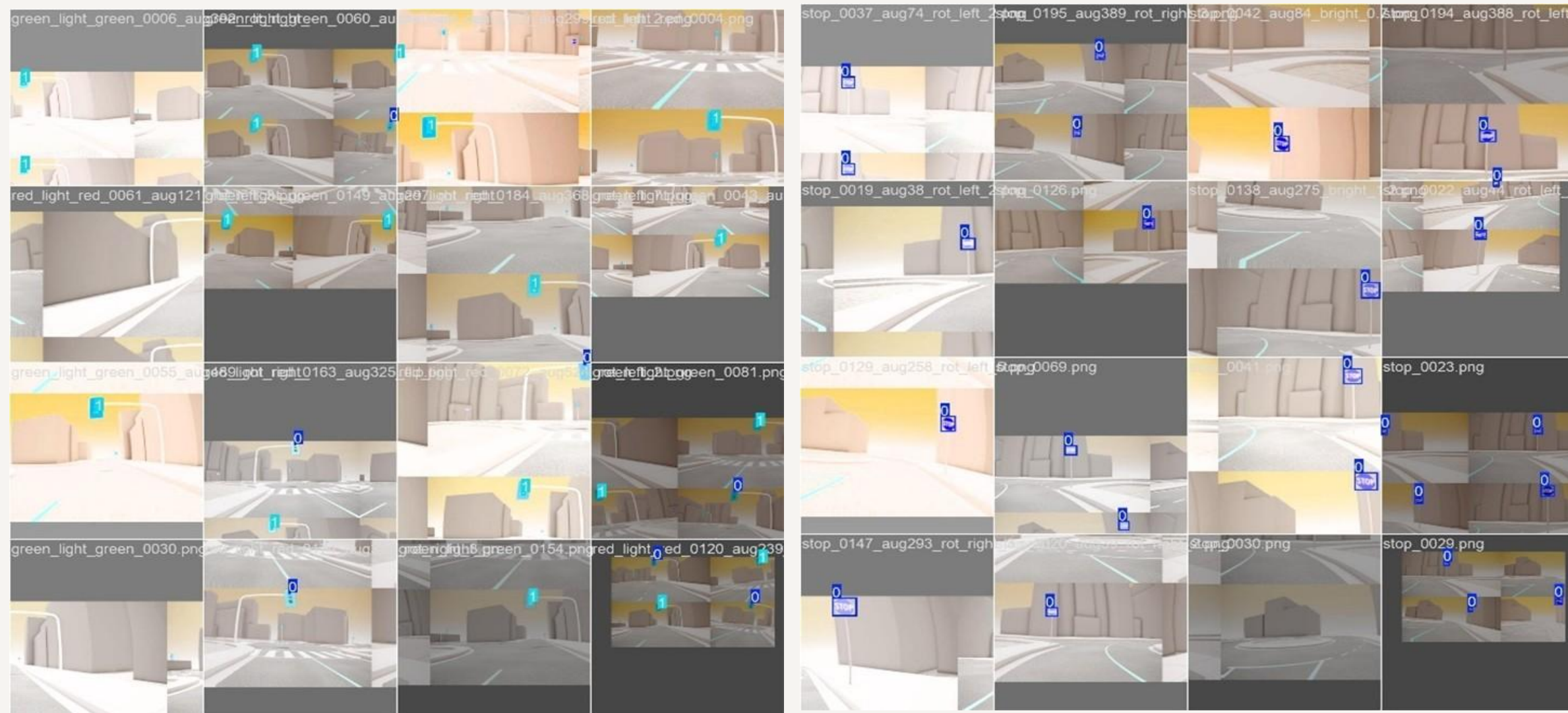


Fig. 4 Simulation training data examples showing, (a) traffic light detection (b) stop sign detection with YOLO bounding box annotations from QLABs virtual environment.

Methodology - Hardware Phase (QCar)

- Object detection: Unified YOLO model (5 classes: red, yellow, green, stop sign, no-right-turn sign)
- Detection dataset: 3,802 instances from varied environments, lighting (0.3–1.5m range), 70/15/15 split
- Lane segmentation: 1,847 frames annotated with 3 classes (background, yellow, white roundabout)
- Data augmentation: rotation, flipping, brightness, blur, noise



Fig.5 QCar platform with the integrated sensor suite

Methodology - Hardware Phase (QCar)

- Distributed architecture adopted due to Jetson TX2 computational limits
- QCar (control client): acquires sensors and executes motor commands
- Laptop(perception server): ENet + YOLO inference, sensor fusion, control generation
- Communication: TCP/IP over Wi-Fi using Quanser BasicStream

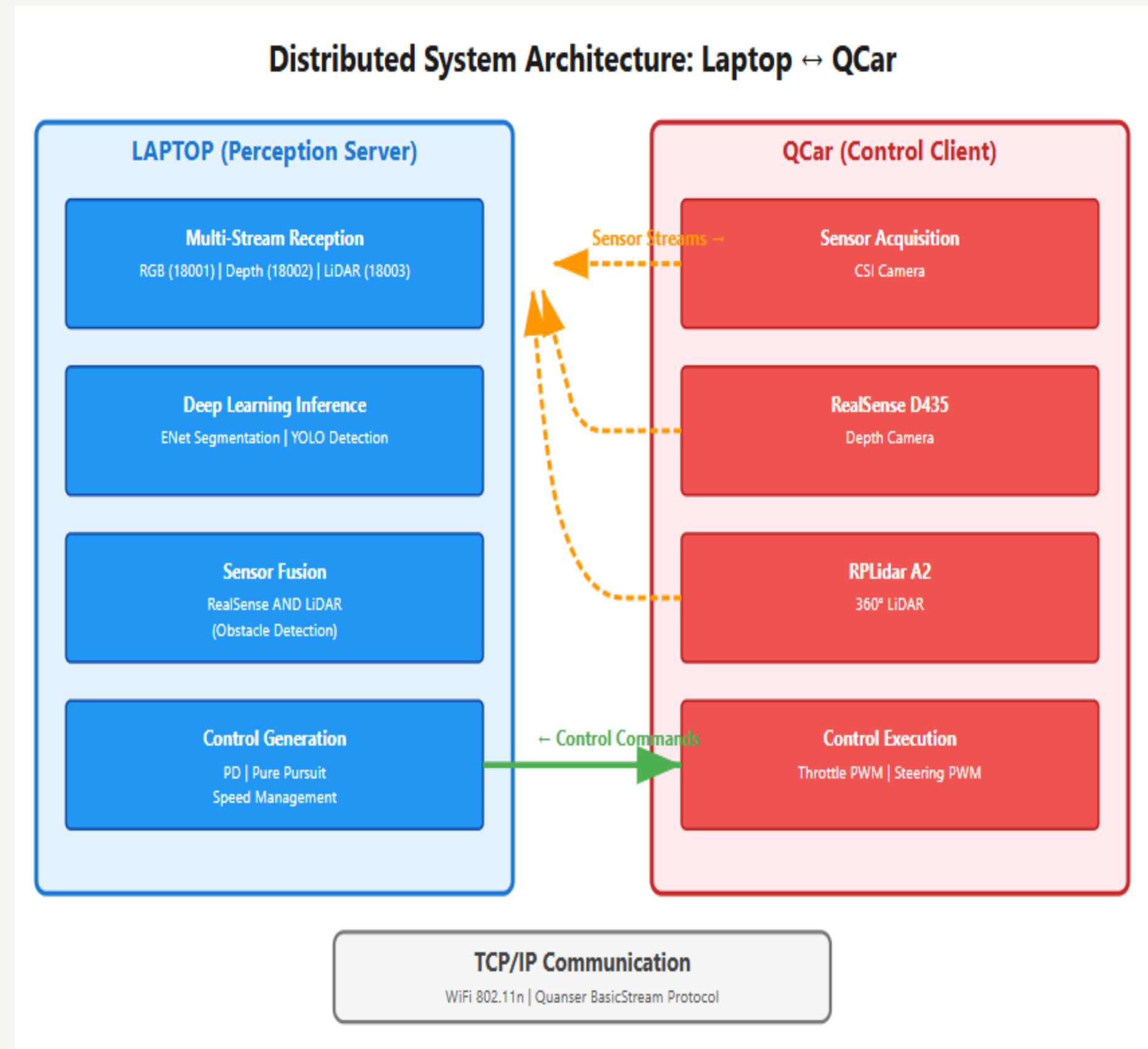


Fig. 6 Distributed System Architecture

Methodology - Sensor Fusion

- Multi-sensor approach: Intel RealSense D435 (depth)
+ RPLiDAR A2 (360° laser scan)
- RealSense: active IR sensing, 0.2–0.4m effective range, centre 60% ROI
- RPLiDAR: 0.8m threshold, strong detection requires 20+ points
- AND-logic fusion: both sensors must agree before triggering emergency braking
- Prioritises zero false positives over maximum detection range
- Consensus zone: 0.2–0.6m where both sensors operate reliably

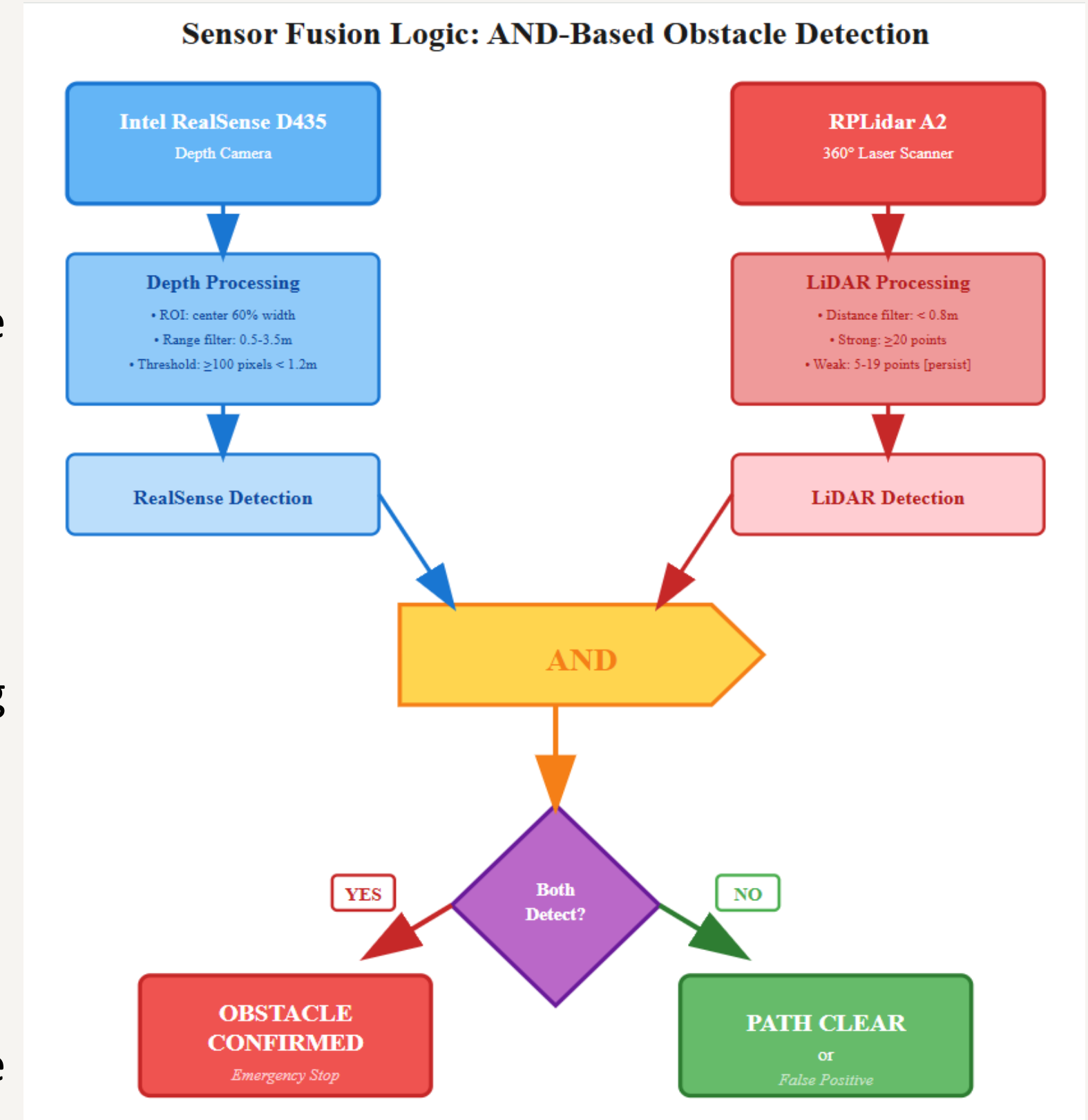


Fig.7 AND-based sensor fusion logic

Results - Simulation Object Detection & lane following

➤ Traffic lights: 99.44% mAP@0.5 (100 epochs, 38 min)

➤ Classification accuracy: red 84%, green 100%, stop

100%

➤ Primary error: red light background false positives

(16%)

➤ Stop signs: 99.50% mAP@0.5 (82 epochs, 12 min)

➤ Proximity-based detection activation to reduce false positives



Fig. 8 Traffic light detection



Fig. 9 Stop sign detection

Results - Simulation Object Detection & lane following

- Lane detection: HSV-based segmentation identifying yellow centrelines and white boundaries
- Path computation: midpoint calculation between detected lane markings
- Traffic compliance: stop at red light, proceed on green, 2 seconds stop at stop signs
- Controlled simulation environment validated algorithms before hardware deployment



Results - Hardware Object Detection

- Unified 5-class YOLO model: red, yellow, green, stop sign, no-right-turn
- Performance metrics: 92.87% mAP@0.5
- Training: 3,802 instances, 100 epochs, 186 minutes
- Steady convergence: mAP improved from 63.6% to 92.9% over 100 epochs
- Background false positives highest for traffic lights (red 42%, yellow 29%, green 18%)

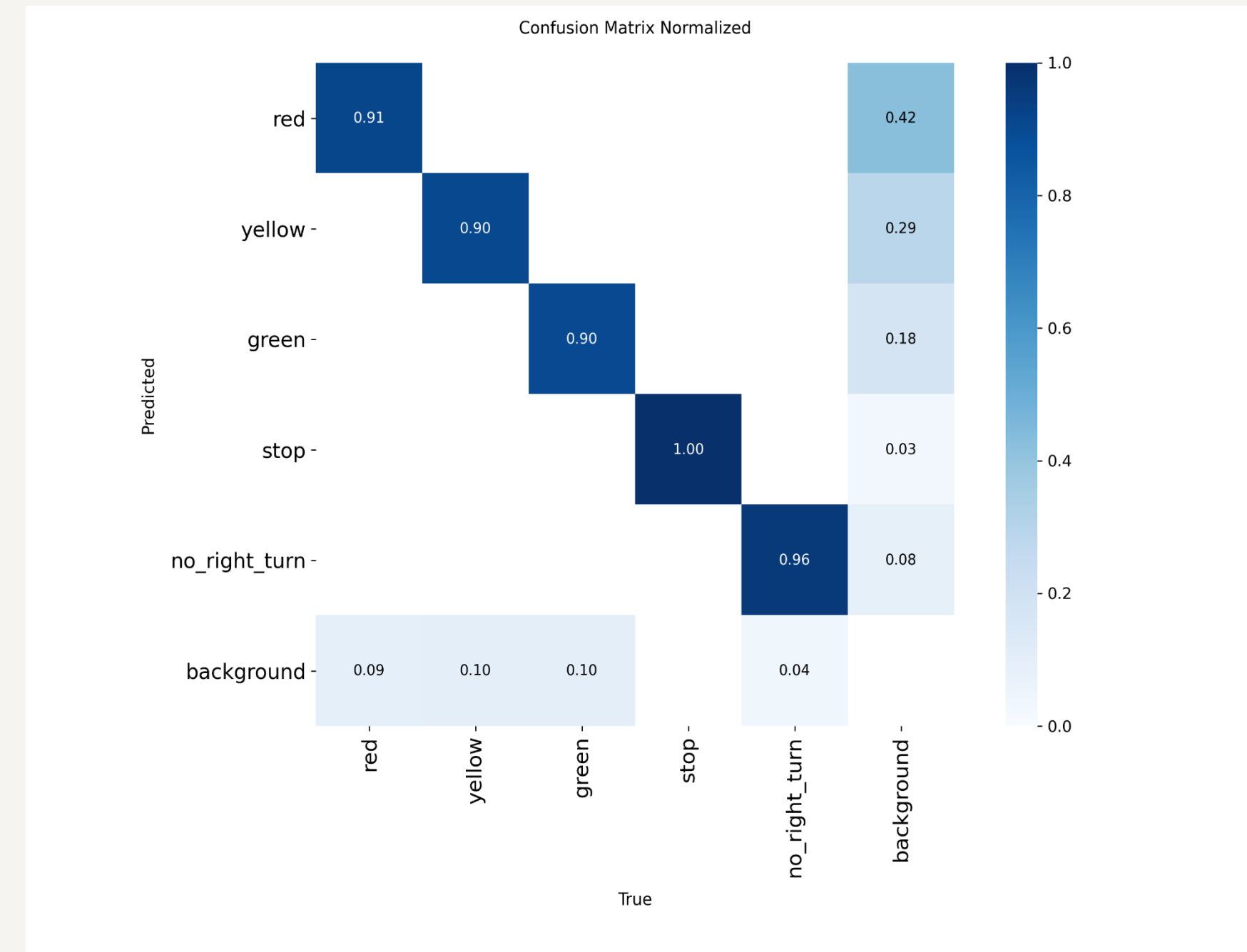


Fig.10 Normalised Confusion Matrix

Results – Hardware Robustness Testing (Detection)

Traffic Lights:

- Occlusion: red maintained up to 50%, yellow/green failed at 50%
- Distance: reliable across 0.4–1.0m range
- Lighting: stable in bright/normal/dim (>0.72 conf), degraded in dark (0.32-0.48)

Traffic Signs:

- Occlusion: stop sign and no-right-turn maintained up to 50%, failed at 75%
- Distance: stop sign reliable 0.2–0.8m, no-right-turn failed beyond 0.6m
- Lighting: signs more robust than lights in dark conditions (stop 0.76, no-turn 0.78)

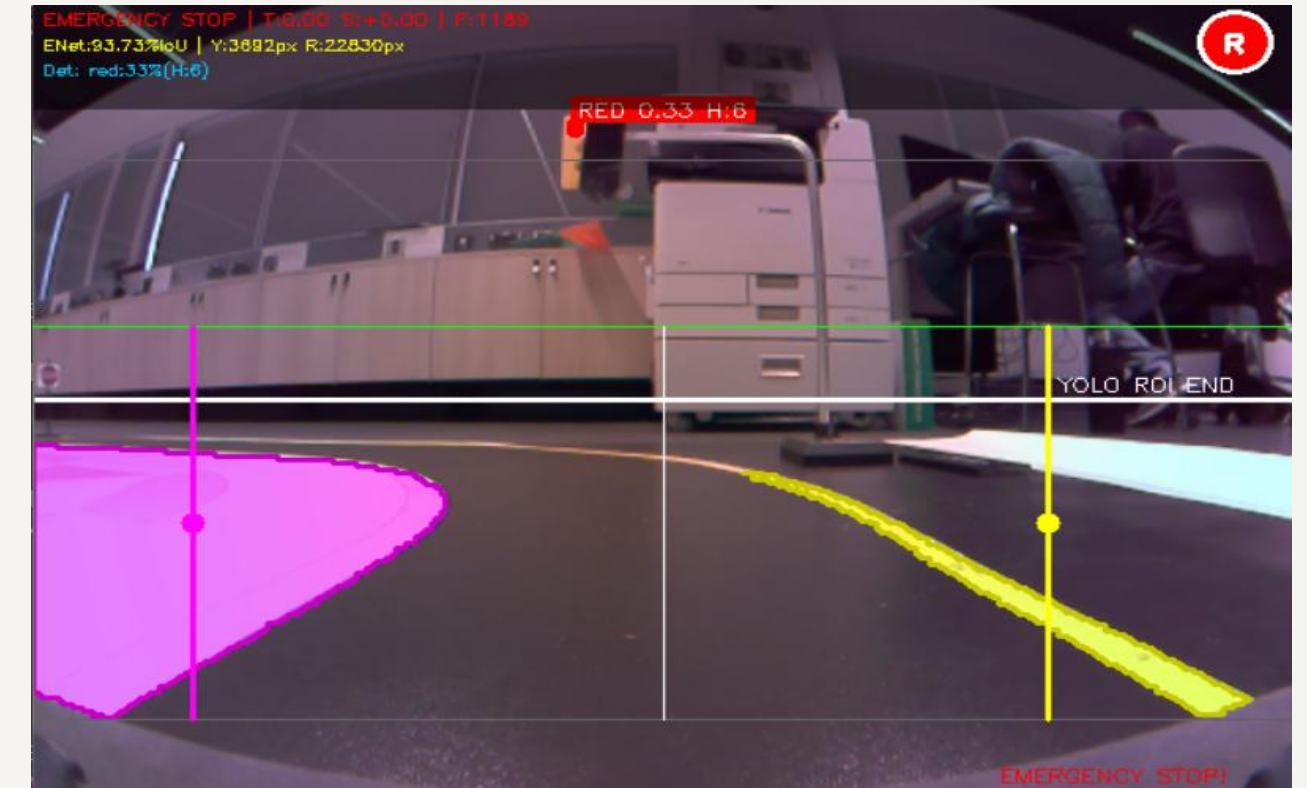


Fig. 11 Red light occlusion test at 50%

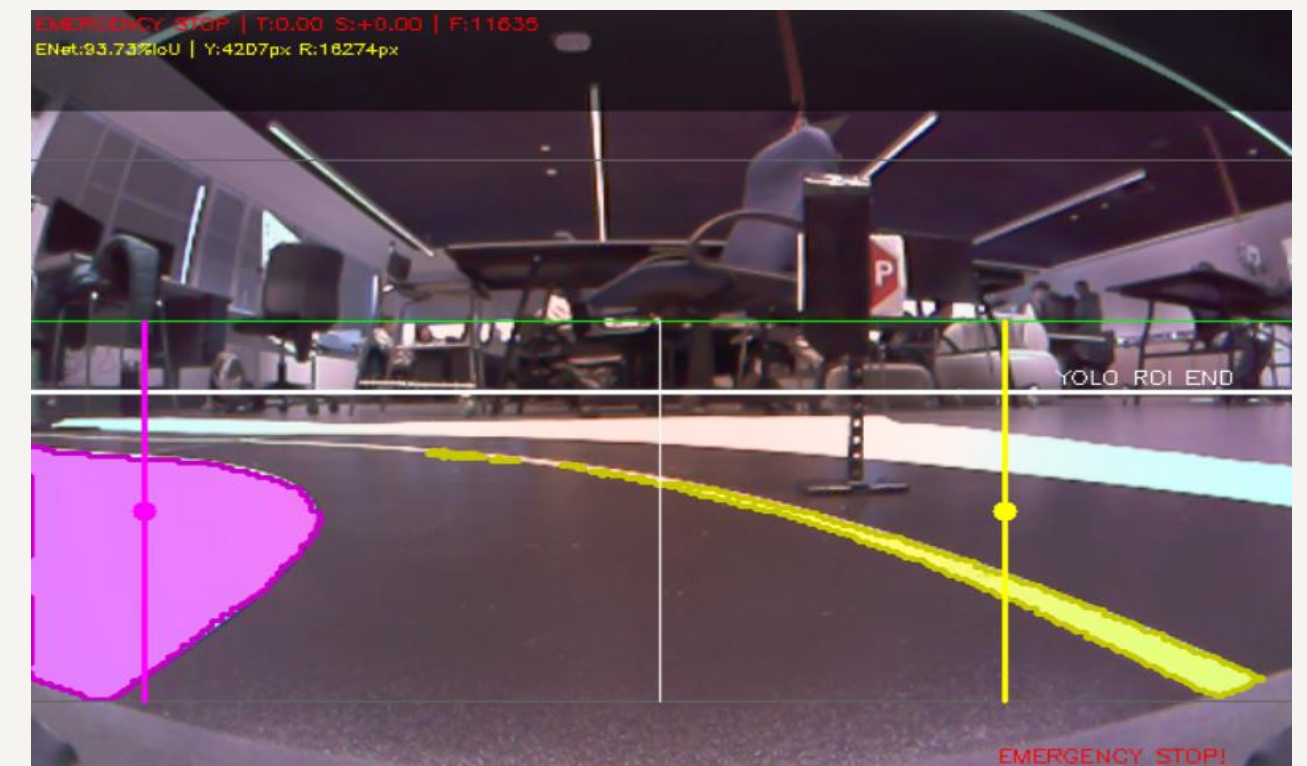


Fig. 12 Stop sign occlusion test at 75%

Results – Hardware Robustness Testing (Segmentation)

- Both U-Net and ENet maintained detection up to 90% occlusion
- Dark conditions: ENet 13× better than U-Net (728 vs 54 roundabout pixels)
- Yellow lane detection more robust than white/roundabout markings
- Lighting: stable detection for across normal/bright/dim, minimal variation



Fig.13 Enet tested at 90% occlusion

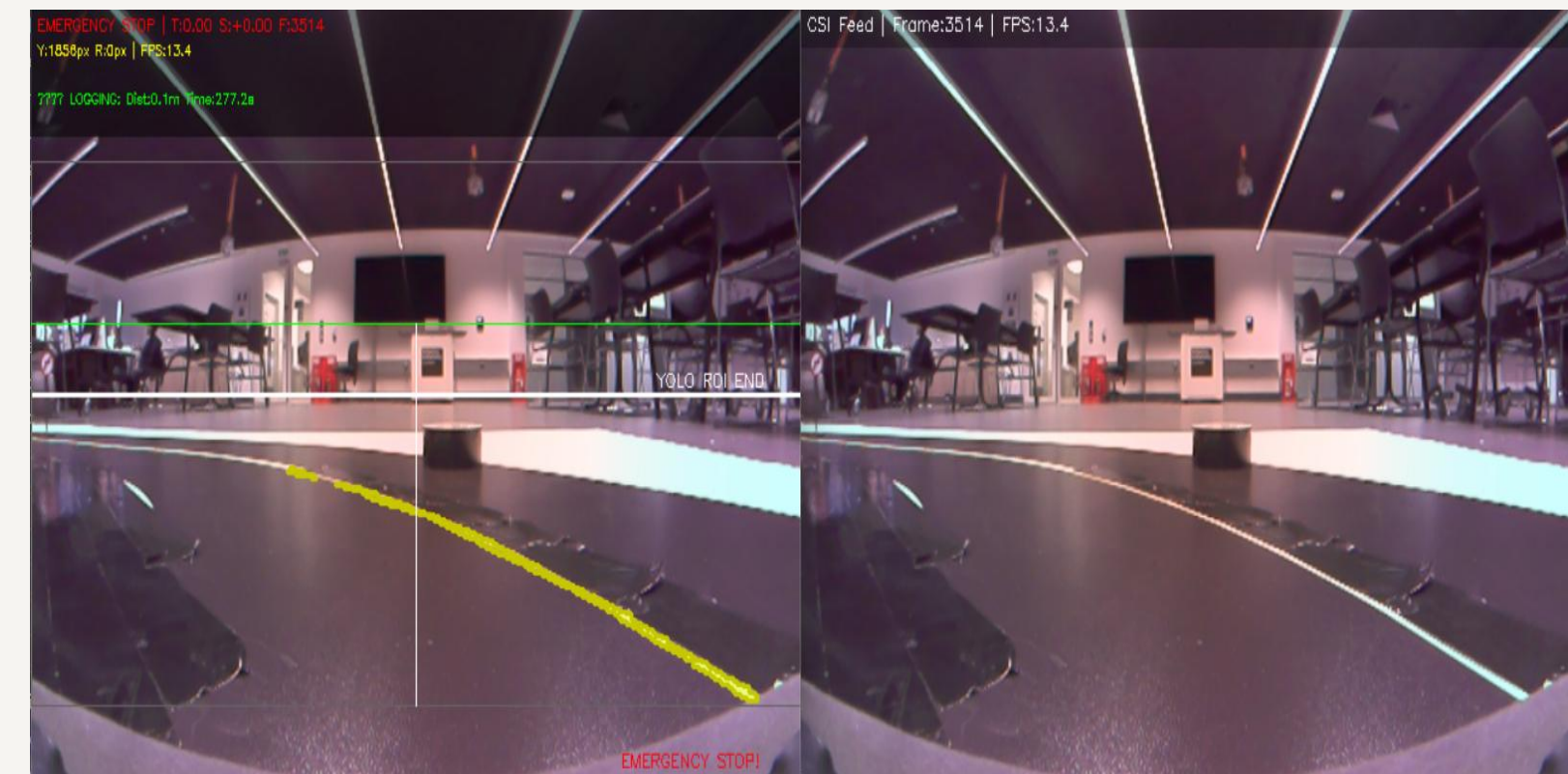
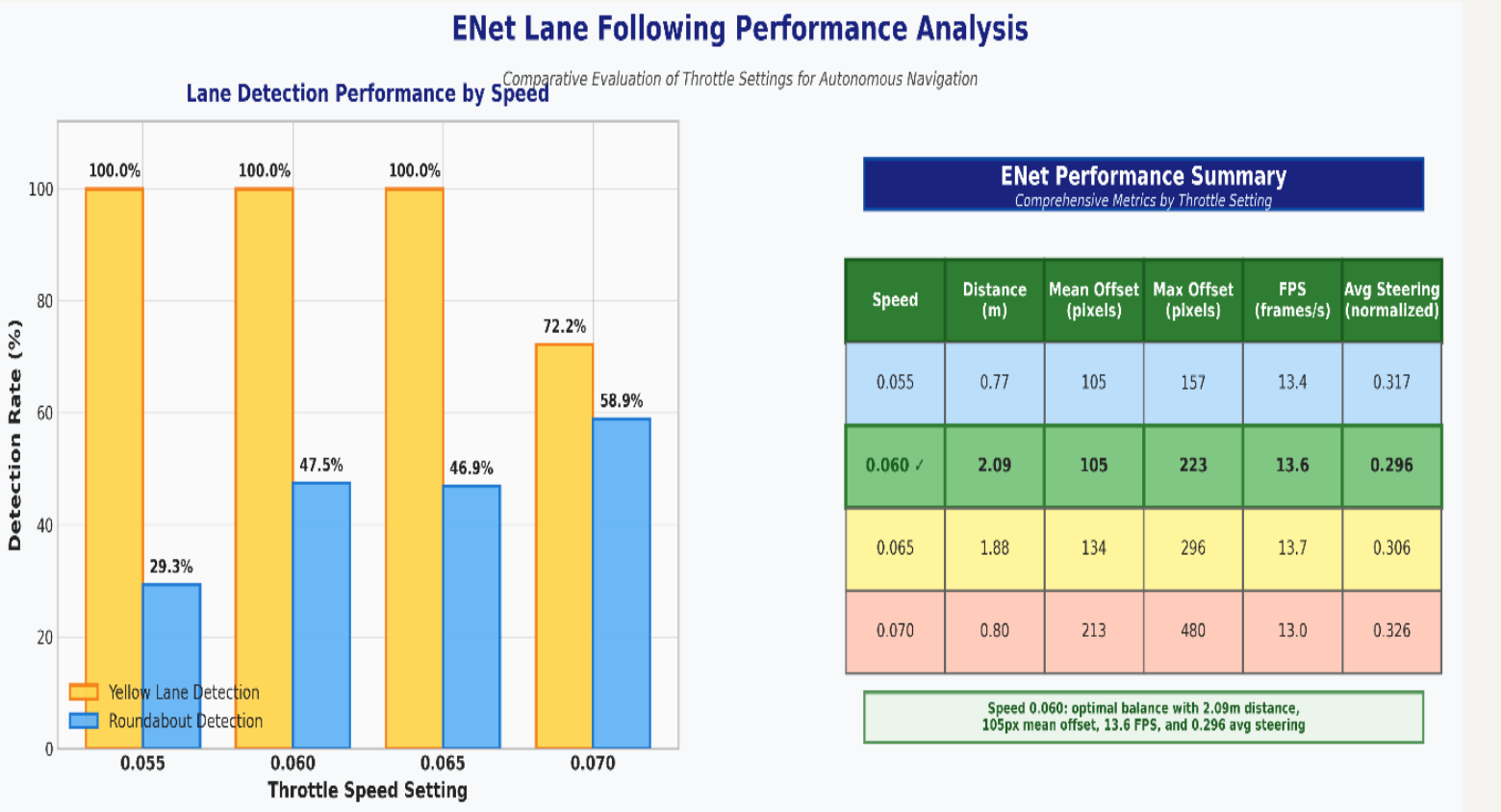
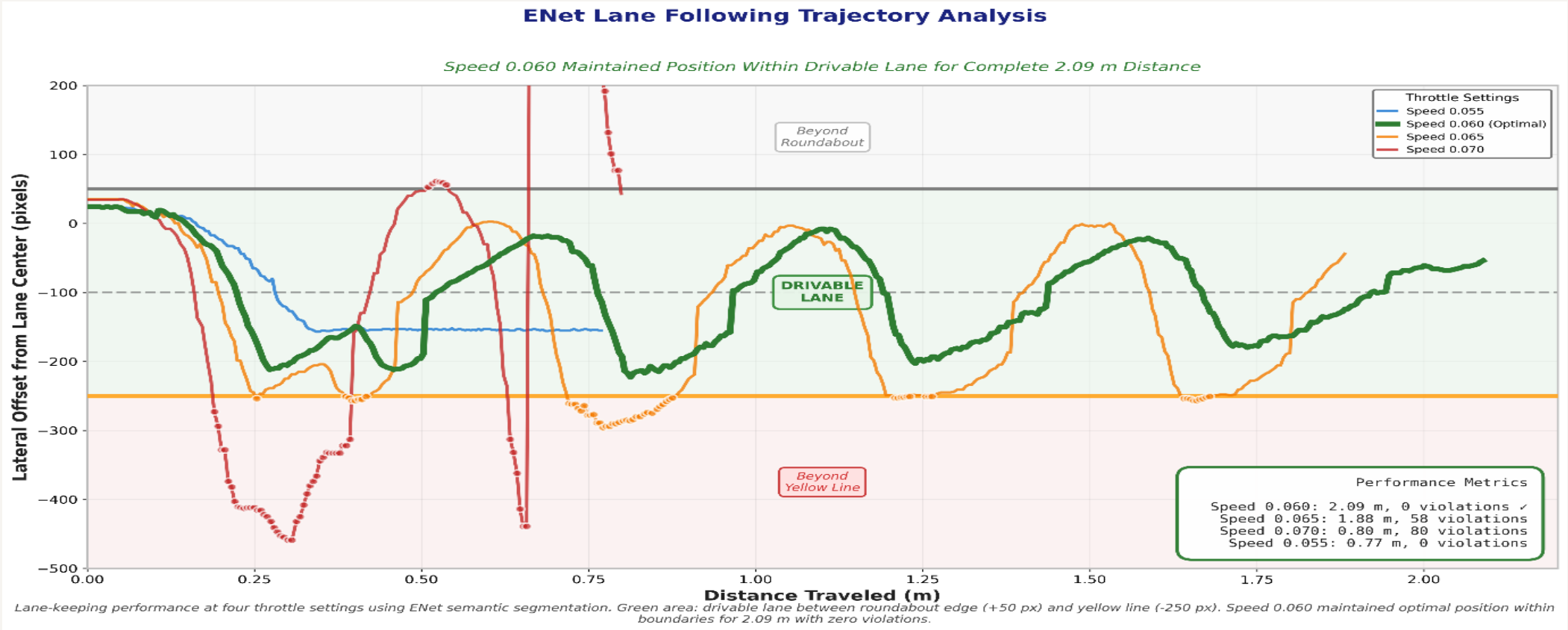
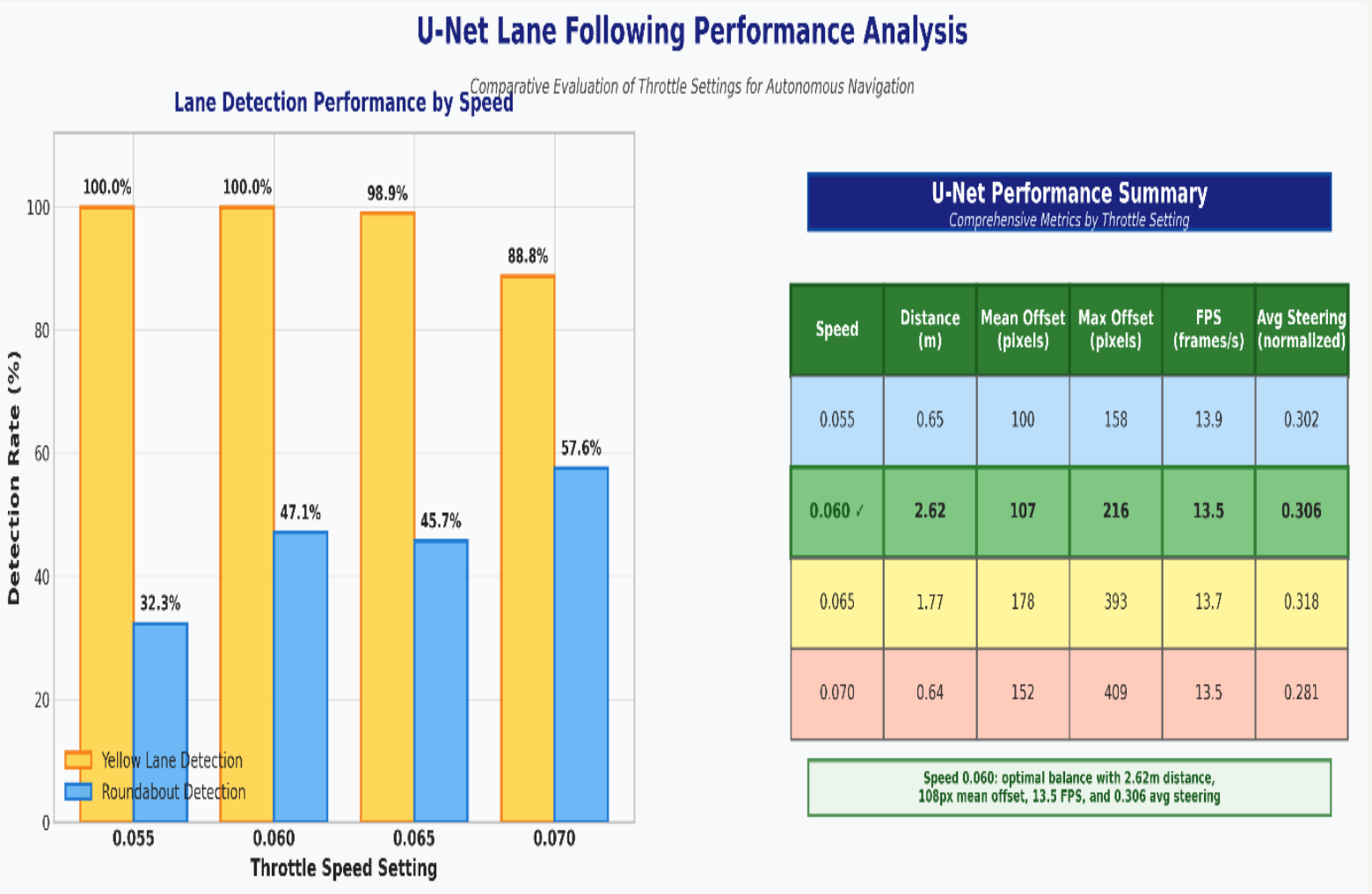
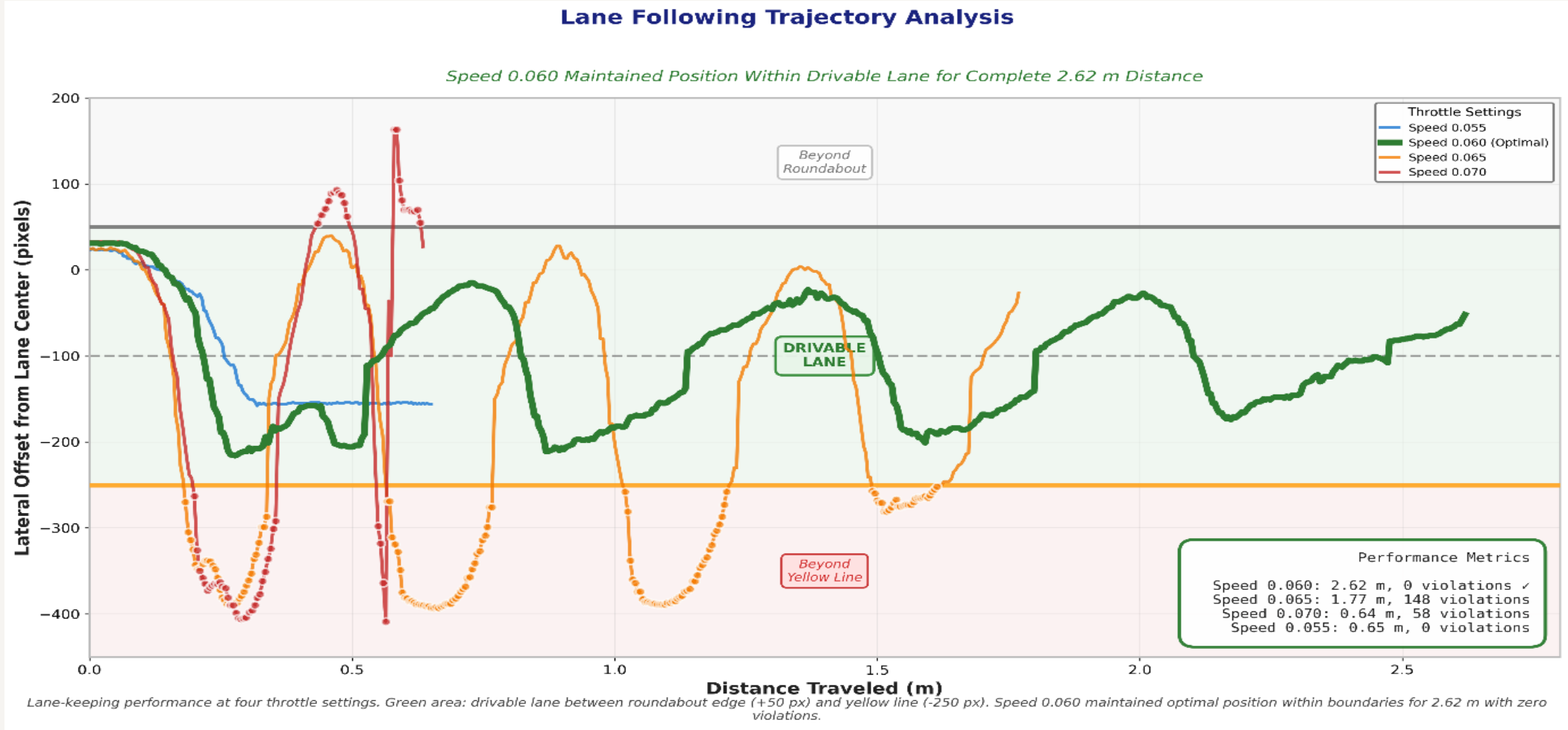
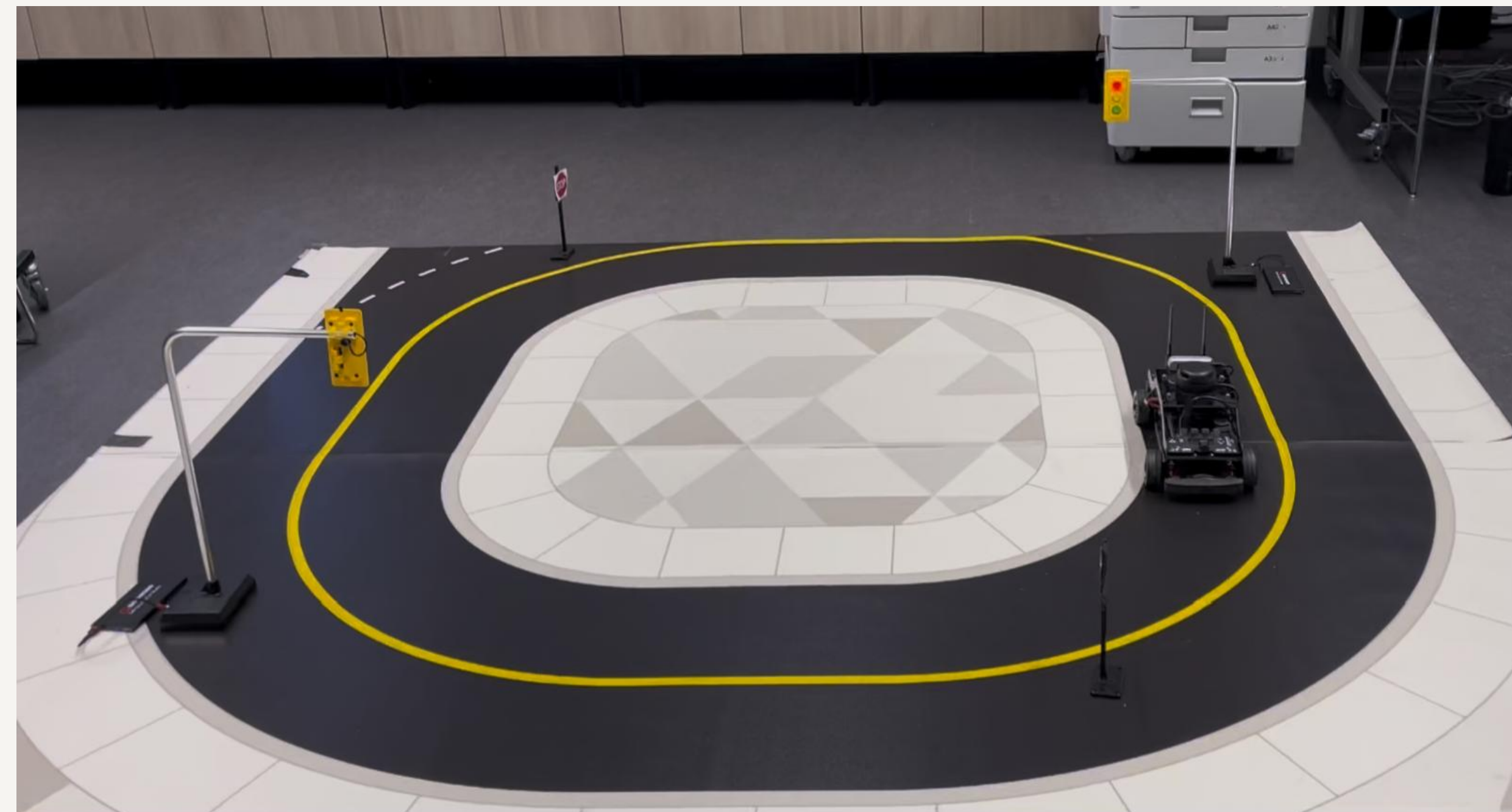


Fig.14 U-net tested at 90% occlusion

Results – Hardware Robustness Testing (lane following)



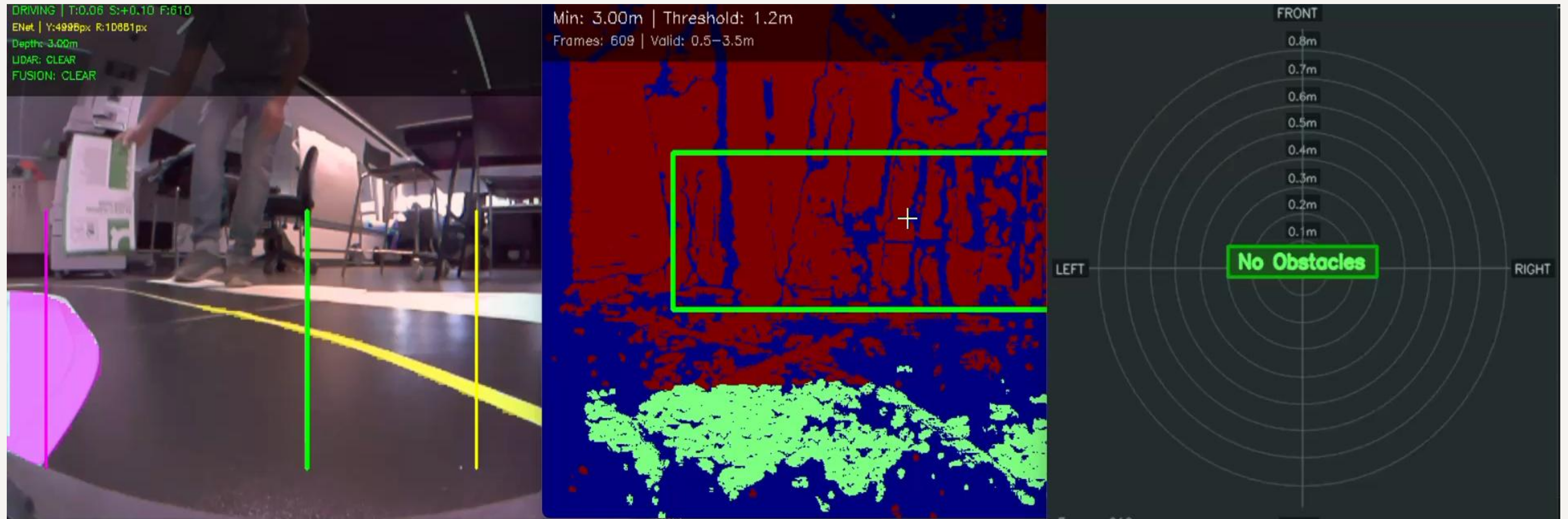
Results - Hardware Object Detection & lane following



Results – Sensor Fusion Validation



Results – Sensor Fusion Validation



- Both sensors lighting independent: identical performance across all conditions
- Depth ROI filtering: ignores car body ($<0.5\text{m}$) and distant noise ($>3.5\text{m}$)



Conclusion

- Feasibility: Vision-based autonomous navigation is practical on educational platforms, achieving real-time closed-loop operation
 - Reality gap quantified: 6.57 percentage point sim-to-real drop
 - Efficiency vs accuracy: ENet achieved 93.73% mIoU with 87× fewer parameters than U-Net, enabling embedded deployment
 - Safety layer: AND-logic sensor fusion provided zero false positives during hardware trials
 - Key limitation: Low-light performance is a major weakness, with system failure under very dark conditions
 - Reproducible framework: Complete pipeline documentation enables future researchers to replicate and extend this work
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Thank You



12 January 2026

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